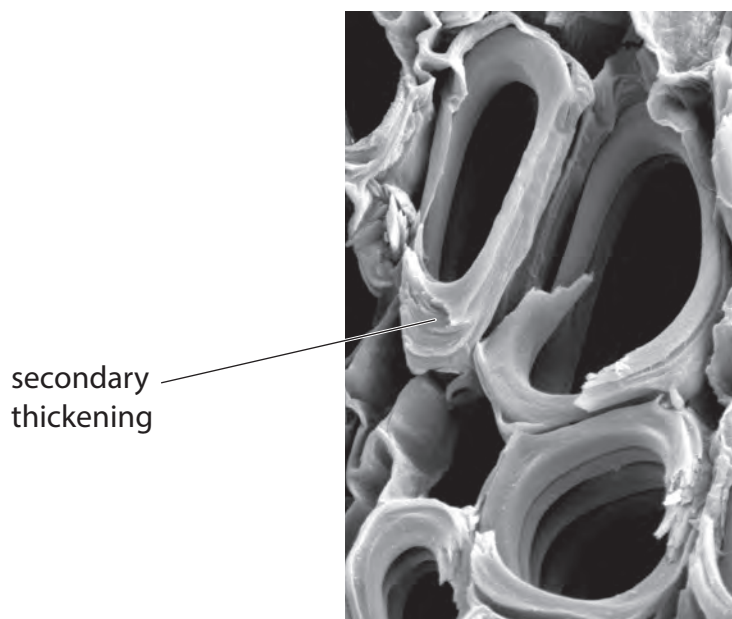


3 The cells in xylem vessels have cellulose cell walls and secondary thickening.

(a) The photograph shows xylem vessels and walls with secondary thickening, as seen using an electron microscope.



(Source: © Steve Gschmeissner / Science Photo Library)

Xylem vessels have cell walls containing cellulose microfibrils.

Describe how the structure and arrangement of cellulose **microfibrils** contribute to their functions in the cell walls of xylem vessels.

(3)

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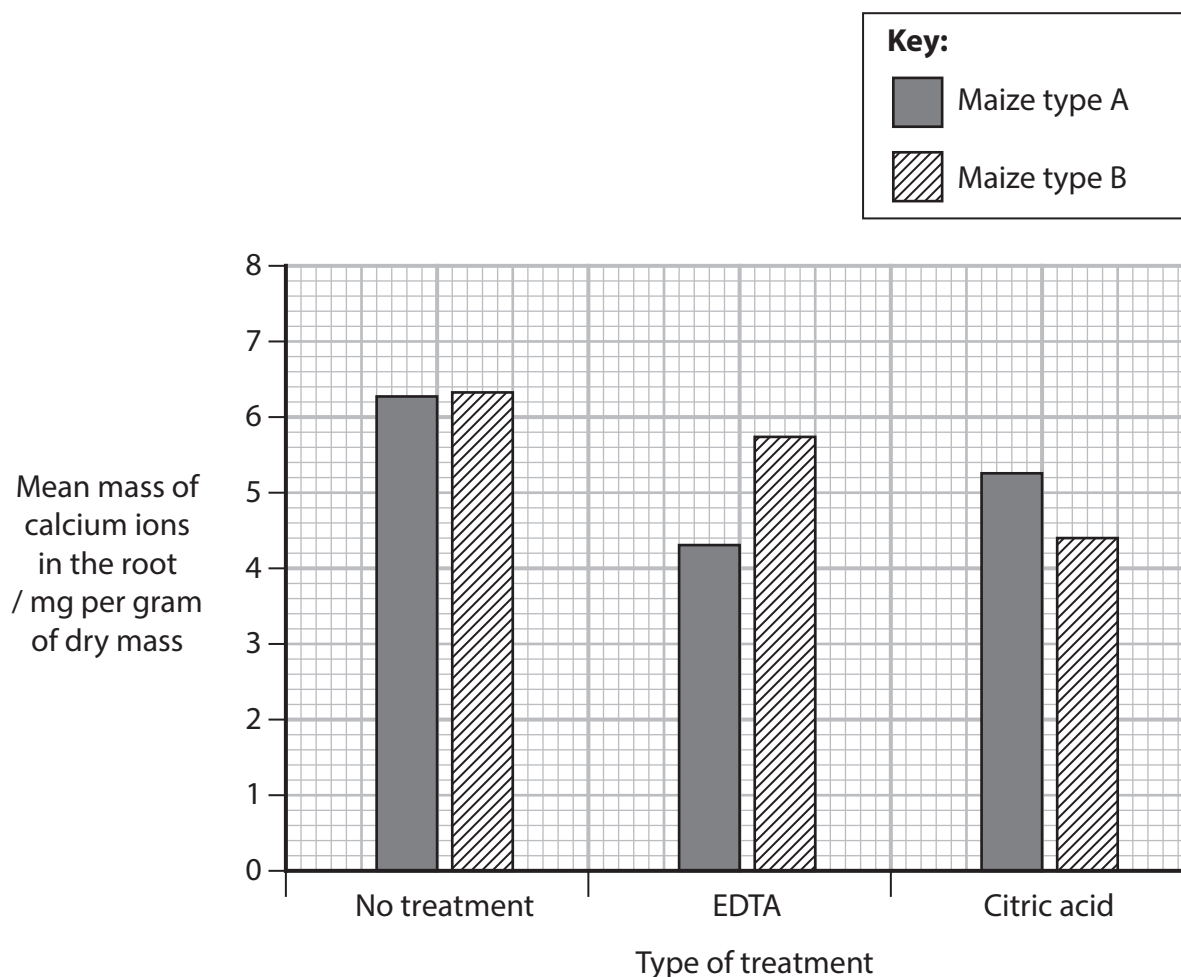
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- (b) Chemicals in the soil can affect the absorption of inorganic ions.

Xylem vessels transport calcium ions from roots to other parts of a plant.

The effect of adding two chemicals, EDTA and citric acid, to the soil was investigated.

The graph shows the mean mass of calcium ions in the roots of two types of maize used in this investigation.



- (i) Give **two** conclusions for the effect of these treatments on the calcium ion content in the roots of these two types of maize.

(2)

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- (ii) Explain the effect EDTA would have on new cell walls in the roots of type A maize plants.

(2)

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(Total for Question 3 = 7 marks)

